# Data Acquisition

## Digital Forensics: Acquire

### Collecting, Gathering, and Preservation

**NSSECU3 | Digital Forensics**

| | |
|---|---|
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# Overview

1. Open-source platforms (OSPs)
   - Preparing OSPs
   - Types
   - Acquisition OSPs
   - Analysis OSPs
2. Data Acquisition using OSPs
   - What & Imaging vs Copying and Why
   - Important Reminders
   - Order of Volatility
   - Types of Data Acquisition
   - Types of Data Acquired
   - Tools and Platforms for Data Acquisition

# Open-Source Platforms

OSPs – Preparing, Types, For Acquisition and Analysis

# Preparing for OSPs

Open-Source Platform/Tool Preparations:

1. Virtualized Machines (VMs)
2. OSPs List and Downloaded
3. Installed latest & updated OSPs
4. Check if OSPs are running as they should
5. Snapshot VM – Name it as “Ready Tools”

# Types of OSPs

Open-Source Platform Types:

1. Self Built OSPs – Self created platforms
2. Publicly Available OSPs – the usual OSPs
3. *Free-Trial Commercialized Platforms – Free platforms for a certain time; mostly good for testing, comparing and not for reliable usage

# Open-Source DF Acquisition Tools

DF Acquisition Platforms (to download and install):

1. SIFT Workstation – https://www.sans.org/tools/sift-workstation/
2. FTK Imager – https://www.exterro.com/ftk-imager#:~:text=FTK%C2%AE%20Imager%20is%20a,(FTK%C2%AE)%20is%20warranted.
3. WireShark – https://www.wireshark.org/download.html

# Open-Source DF Analysis Tools

DF Analysis Platforms (to download and install):

1. SIFT Workstation – https://www.sans.org/tools/sift-workstation/
2. FTK Imager – https://www.exterro.com/ftk-imager#:~:text=FTK%C2%AE%20Imager%20is%20a,(FTK%C2%AE)%20is%20warranted.
3. WireShark – https://www.wireshark.org/download.html

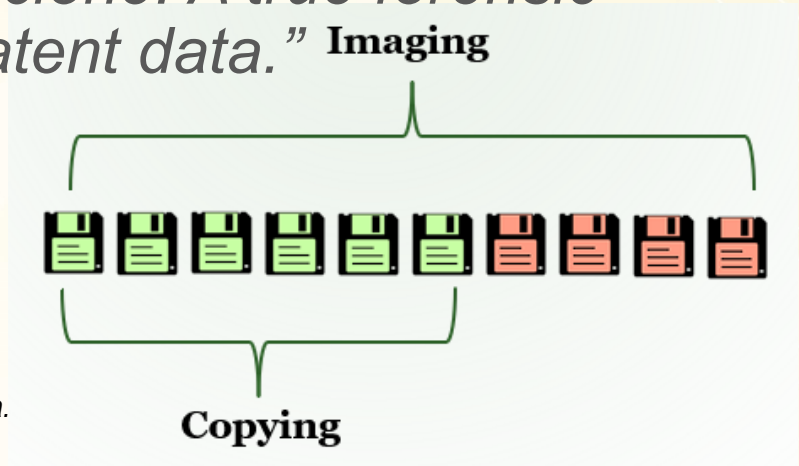
# Data Acquisition

Acquiring and More

# What is Acquisition?

Acquisition is making a forensic copy of evidence, which could be any type of media (Hard disk drive, USB, CD/DVD, etc.).

*"A forensic image of a hard drive captures everything on the hard drive, from the physical beginning to the physical end. Performing a "copy and paste" via the operating system is not the same as a forensic clone. A true forensic image captures both the active and latent data."*




https://www.sciencedirect.com/topics/computer-science/forensic-clone#:~:text=Digital%20Forensics&text=A%20forensic%20image%20of%20a,the%20active%20and%20latent%20data.

# Why acquire?

- Evidence <u>might be damaged while working on it</u>

- Evidence <u>might be in use</u>, or the client cannot afford to let the investigator keep the machine until the end of investigation.
  – *Imagine bringing home a whole server for analysis*

- Multiple investigators **cannot work** on <u>a single evidence</u>.
  – *One might be working on registry and the other might work on machine logs*

# Important Reminders

- **Do not work** on the original machine while investigating.

- **Do not work** on the primary image that was acquired.
  – *If there are changes on the primary image, the whole process of investigation might be invalidated and inadmissible in court.*

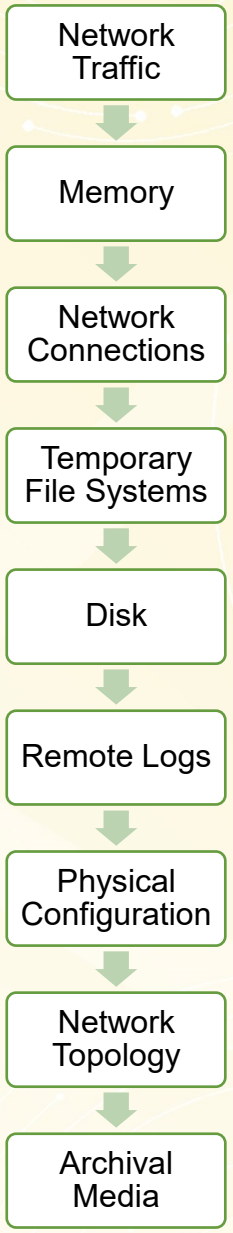
- Investigator **should always** consider the order of volatility when acquiring evidence from the machine

# Order of Volatility

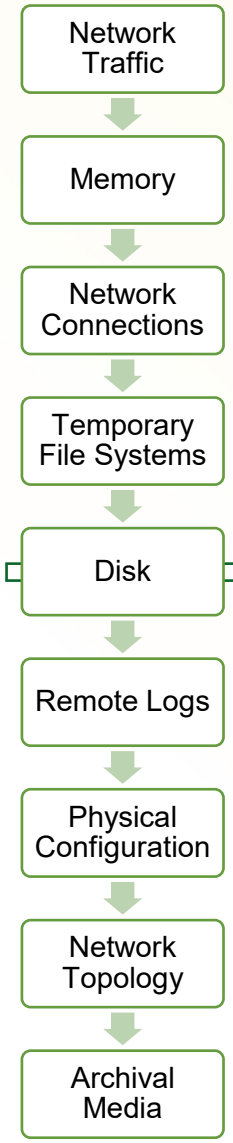
## Data Acquisition Complexity: Order of Volatility

The Internet Engineering Task Force (IETF) has put together a document titled Guidelines for Evidence Collection and Archiving (RFC 3227) that addresses the order of volatility of digital evidence, as follows:

1. Registers and cache
2. Routing table, ARP cache, process table, kernel statistics, memory (RAM)
3. Temporary filesystems
4. Disk
5. Remote logging and monitoring data
6. Physical configuration, network topology
7. Archival media



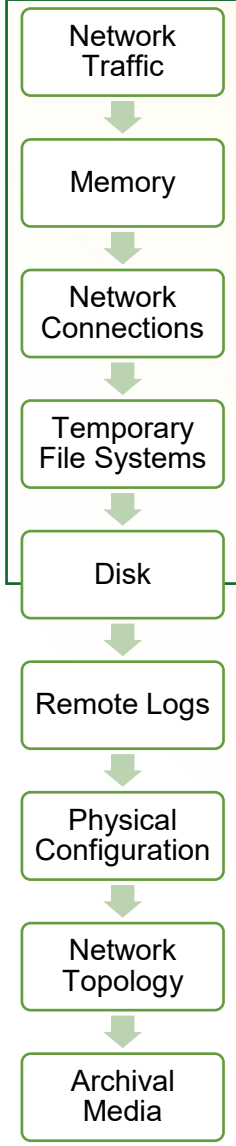
# Order of Volatility: Types of DA




- Dynamic (Live) Acquisition
  - Gathering volatile data while the device is still running

- Static (Offline) Acquisition
  - Gathering nonvolatile data. In other words, gathering the data that remains intact after the system's reboots or goes down.

# Order of Volatility: Types of DA




- Dynamic (Live) Acquisition
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# Order of Volatility: Types of DA

- Network Traffic
- Memory
- Network Connections
- Temporary File Systems
- Disk
- Remote Logs
- Physical Configuration
- Network Topology
- Archival Media

- Dynamic (Live) Acquisition
  - Gathering volatile data while the device is still running

- Static (Offline) Acquisition
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# Order of Volatility: Network Traffic

Network Traffic

Memory

Temporary File Systems

Disk

Remote Logs

- Acquisition
  - WireShark
  - tcpdump
- Analysis
  - WireShark
  - Network Miner

- File Format(s)
  - ***.PCAP** – Packet Capture
  - ***.PCAPNG** – Packet Capture Next Generation

# Order of Volatility: Memory

Network Traffic

Memory

Temporary File Systems

Disk

Remote Logs

- Acquisition
  - Memoryze
  - DumpIT
  - FTK Imager
- Analysis
  - Volatility (in SIFT Workstation)

- File Format(s)
  - ***.DMP/*.MEM** – RAW memory dump; Extracted from a live environment
  - ***.CRASH** – Crash Dump Information gathered by the operating system
  - **hiberfil.sys** – Windows hibernation files; A saved snapshot that your operating system can return to after hibernating
  - **pagefile.sys** – Page File; this is a file that stores similar information that is stored in your system RAM
  - ***.VMEM** – VMWare Snapshot; This is a snapshot of a virtual machine, which saves its state as it was at the exact moment that the snapshot was generated

# Order of Volatility: Disk

Network Traffic

Memory

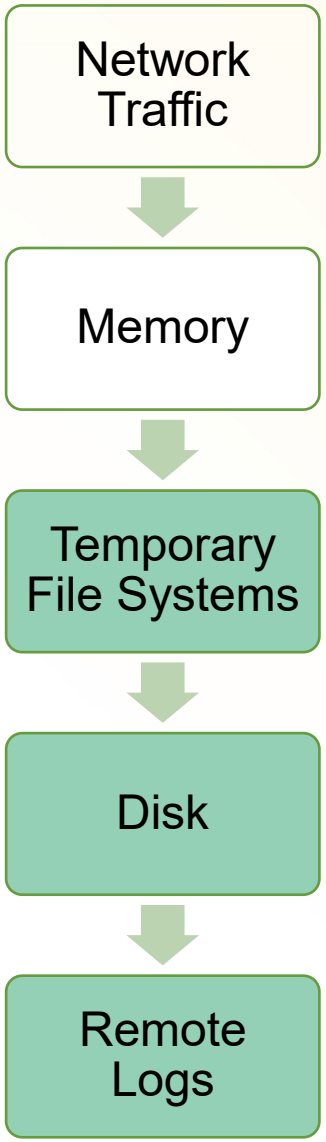
Temporary File Systems

Disk

Remote Logs

- Acquisition
  - FTK Imager
  - dd (in SIFT Workstation)
- Analysis
  - FTK Imager
  - Autopsy
  - Sleuth Kit (in SIFT Workstation)
- File Format(s)
  - ***.E01** – EnCase/X-Ways/Evidence Image File
  - ***.000/*.RAW** – Raw Image File (Single)
  - ***.dd** – Linux DD (Disk/Disk Duplicator)
  - ***.*0*** – Segmented Image Files (e.g., e01, e02, or 001, 002)

# Order of Volatility: Disk-TMP-LOGS




- Temporary File Systems and Logs
  - Can be captured Live and Offline.
  - From the Image File – can be carved (data carving)
  - Can be copied manually instead of using a tool or platform
  - Further discussions will be on Artifacts (Windows mostly)

# Other Open-Source Platforms

## Notable OSPs

- Autopsy
- CyberChef
- Network Miner

Live Response and Analysis OSPs:

- Windows SysInternals
- HashMyFiles

Sysinternals Suite

Article • 04/12/2023 • 8 contributors

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